Q1)

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 2 \end{bmatrix}$$

a) Determine the inverse matrix.

Augment the matrix with the identity matrix

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array}\right] \quad \begin{array}{l} R_3' = R_3 - R_1 \\ R_2' = R_2/2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & -1 & 0 \\ 0 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & -1 & 1/2 & 1 \end{array}\right] \quad \begin{array}{l} R_1' = R_1 - 2R_2 \\ R_3' = R_3 + R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & -1 & 0 \\ 0 & 1 & -1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & -2 & 1 & 2 \end{array}\right] \quad R_3' = 2R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & -3 & -4 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & -2 & 1 & 2 \end{array}\right] \quad \begin{array}{l} R_1' = R_1 - 2R_3 \\ R_2' = R_2 + \frac{R_3}{2} \end{array}$$

$$A^{-1} = \begin{bmatrix} 5 & -3 & -4 \\ -1 & 1 & 1 \\ -2 & 1 & 2 \end{bmatrix}$$

b) solution of system of linear equations

System of equation in matrix form

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$A \qquad X = b$$

$$AX = b$$

$$X = A^{-1} b = \begin{bmatrix} 5 & -3 & -4 \\ -1 & 1 & 1 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$X = \begin{bmatrix} 5-6-12 \\ -1+2+3 \\ -2+2+6 \end{bmatrix} = \begin{bmatrix} -13 \\ 4 \\ 6 \end{bmatrix}$$

$$\boxed{\begin{aligned} x_1 &= -13 \\ x_2 &= 4 \\ x_3 &= 6 \end{aligned}}$$

I

nullspace of $\begin{bmatrix} -\lambda-3 & 13 \\ -1 & 3-\lambda \end{bmatrix} = \begin{bmatrix} -3+2i & 13 \\ -1 & 3+2i \end{bmatrix}$

$\begin{bmatrix} 1 & -3-2i \\ -1 & 3+2i \end{bmatrix} \quad R_1' = \xrightarrow{\frac{-3-2i}{13}} R_1$

$\begin{bmatrix} 1 & -3-2i \\ 0 & 0 \end{bmatrix} \quad R_2' = R_2 + R_1 \quad \Rightarrow \begin{bmatrix} 1 & -3-2i \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

null space : $x_2 = t \quad x_1 = (3+2i)t$

thus eigenvector

$$\boxed{V = \begin{bmatrix} 3+2i \\ 1 \end{bmatrix}}$$

Q1) a. $0i + 8j + 0k$

b. $0i - \sqrt{5}j + 0k$

c. $0i - \frac{3}{10}j + \frac{2}{5}k$

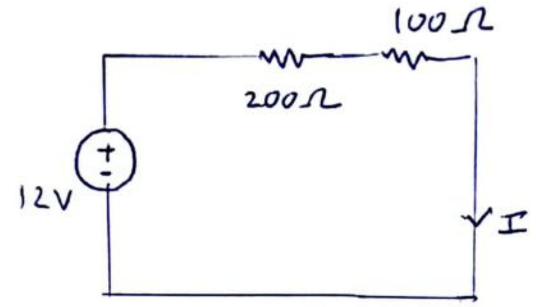
d. $6i - 2j + 3k$

Q2) $r(t) = \left[6 + 3\ln|\sec t|\right]i + \left[5\sin\left(\frac{t}{5}\right) - 7\right]j + \left[5 - \frac{\ln|\tan 2t + \sec 2t|}{2}\right]k$

Q3) $r(t) = \left[2(t+1)^{3/2} - 2\right]i + \left[8 - 8e^{-t}\right]j + \left[\ln|t+1| + 1\right]k$

Q4) $r(t) = \left[2e^{t} - 3t + 3\right]i + \left[-e^{-t} + 3t + 6\right]j + \left[2e^{2t} - 4t + 4\right]k$

Q1) For steady state circuit is simplified as




Using ohm's law $I = \frac{V}{R} = \frac{12}{200+100} = 0.04\,A = 40\,mA$

$\boxed{\text{current through inductor} = 40\,mA}$

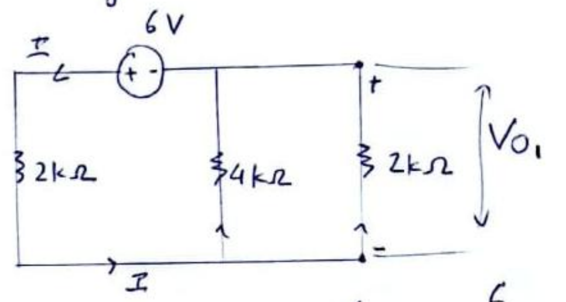
capacitor is in parallel to 200Ω resistor so

$V_c = 200 \times 0.04 = 8V$

$\boxed{\text{Voltage across capacitor} = 8V}$

Q3) superposition theorem for $V_o$

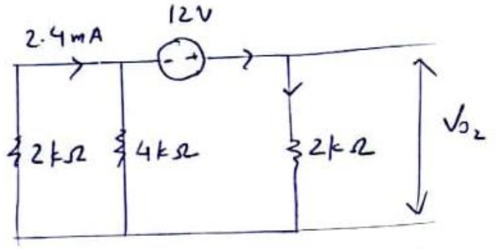
considering 6V battery first




current through 6V battery : $I = \frac{V}{R_{eq}} = \frac{6}{2 + \frac{1}{\frac{1}{4}+\frac{1}{2}}} = 1.8\,mA$

$$V_{o_1} = -1.8\left(\frac{1}{\frac{1}{4}+\frac{1}{2}}\right) = -2.4V$$

consider 12V battery Now




current through battery : $I = \frac{V}{R_{eq}} = \frac{12}{2 + \frac{1}{\frac{1}{4}+\frac{1}{2}}} = 3.6\,mA$

$$V_{o_2} = 3.6 \times 2 = 7.2\,V$$

so $V_o = V_{o_1} + V_{o_2} = 7.2 - 2.4 = 4.8\,V \Rightarrow \boxed{V_o = 4.8V}$

power supplied by 6V battery

$$P = VI = 6(1.8 - 2.4)mA = -3.6\,mW$$

$\boxed{P_{6V} = -3.6\,mV}$ so 6V source is charging

Q5)

5.1 : F-value : $\frac{120.60}{10.10} = 11.94$

$\S$. $DF = k-1 = 3-1 = 2$

Adj SS : $12 \times 10.1 = 121.2$

| | DF | Adj SS | MS | F value | P value |
|---|---|---|---|---|---|
| Factor | 2 | 241.2 | 120.6 | 11.94 | 0.001 |
| Error | 12 | 121.2 | 10.1 | | |
| Total | 14 | 362.4 | | | |

5.2 : $H_0 : \mu_1 = \mu_2 = \mu_3$ , $H_1$ : ($H_0$ is false)

5.3 : F value : $11.94 > 1$ so we reject null hypothesis

Here p-value is $0.001 < 0.05$ confidence interval so we reject null hypothesis